



**THE UNIVERSITY OF QUEENSLAND
SCHOOL OF CIVIL ENGINEERING
CIVL4333 Advanced Concrete Design
Assignment 1**

Question 1

A simply-supported beam with a rectangular cross section as shown in Fig 1 spans 12 m and carries a live load of 32 kN/m. The beam is post-tensioned with a single cable, which consists of ten 7-wire ordinary strands. The tendon profile is parabolic, with no eccentricity at both ends and a maximum eccentricity e of 310 mm at the mid-span. All strands are stressed to 75% of their characteristic breaking loads.

1. Determine the allowable initial prestress force to satisfy the adequacy of section at mid-span at transfer according to Cl8.1.6.2 of AS3600. Considering the assumed immediate losses, is the applied jacking force suitable for design at transfer?
2. What is the additional uniformly distributed load required to achieve decompression at the critical section under service loads?
3. If the beam is subjected to a live load of 32 kN/m, in addition to the self-weight, check whether the beam is fully prestressed under service loads.
4. Calculate the equivalent distributed load due to prestress.

Given the following material properties:

Concrete

Characteristic compressive strength $f_c' = 50$ MPa

Strength at transfer $f_{cp}' = 30$ MPa

Density of concrete (with steel) = 25 kN/m³

Strands

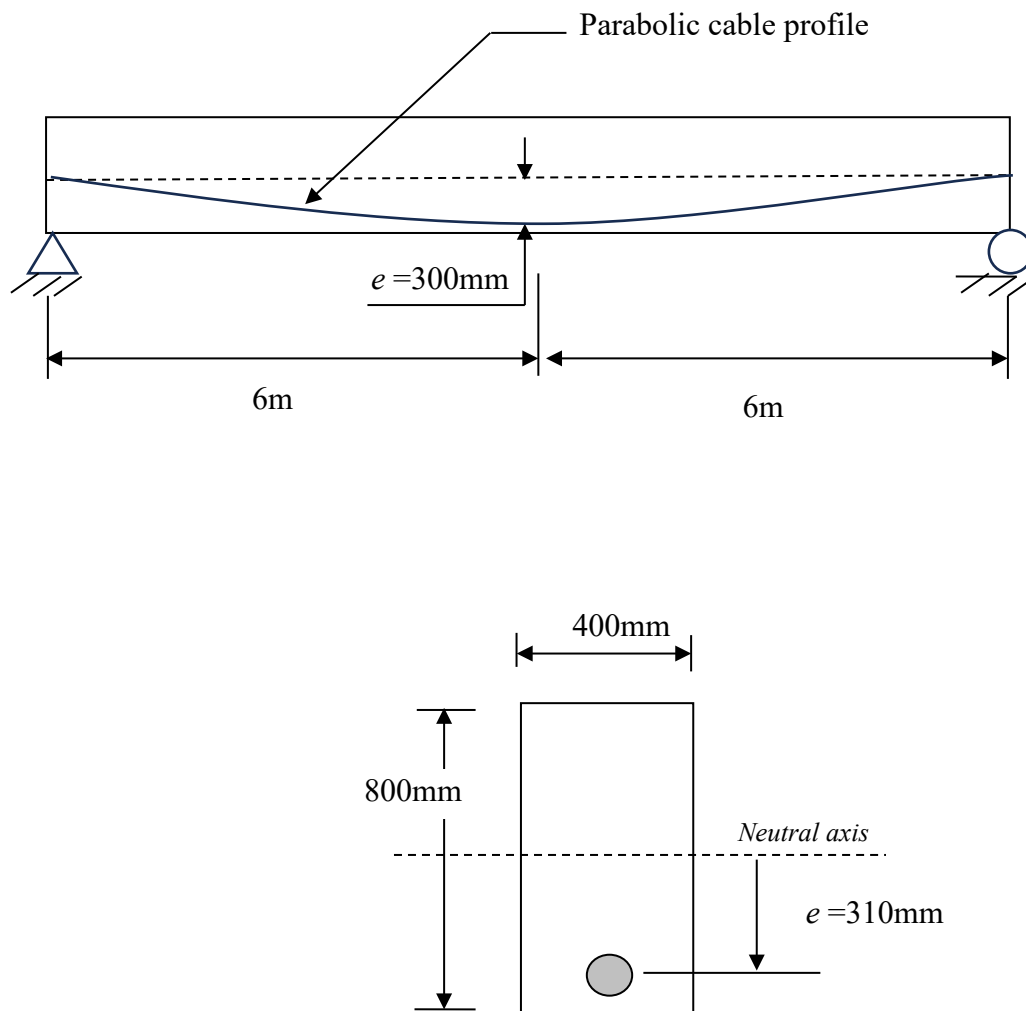
Characteristic breaking load of a 7-wire ordinary strand = 184 kN

Nominal steel area of a 7-wire ordinary strand = 100 mm²

Loss of prestress

Assume immediate losses to be 10%, and the deferred losses after initial prestress to be at 10%, at mid-span.

(See Fig 1 on the next page)



Section at mid span (dimensions in mm)

Fig 1 A post-tensioned prestressed concrete beam

END OF ASSIGNMENT 1